

Schema Activation via Source Attribution: Does Labeling a Passage as AI-Authored Affect Immediate Recall?

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Abstract—A schema is an abstract knowledge structure that organizes knowledge and can shape the processing of new information. Schemas may affect cognitive functions by influencing encoding and recall, which may be utilized to improve learning outcomes. In this study, we investigated whether AI-authored text activates a schema that affects memory recall. We assessed schema activation via a survey split into two parts: a short information passage followed by memory recall questions, and a post-recall section with metacognitive questions. Responses to the metacognitive questions were qualitatively coded into categories for schema analysis, such as credibility, academic quality, and sentiment. Analysis of recall scores revealed that groups exposed to AI-authored sources perform better on immediate recall questions than groups exposed to human-authored sources. Human-authored source groups scored an average of 5.44/9 on the recall portion of the assessment versus the AI-authored source group’s 6.99/9. In addition, the human-authored source groups perceived their source as more credible and academic, while the AI-authored source group saw their source as unreliable, and had lower scores in perceived accuracy. This suggests AI-authored sources activate a low-credibility, high-skepticism schema which increases immediate memory recall.

I. INTRODUCTION

AI-generated content is becoming increasingly common as the technology behind large language models (LLMs) advances, and it has the potential to be particularly transformative in educational contexts. Teachers could use generative AI to aid in creating lesson plans, homework, or exams, and students could employ it as a specialized tutor that is tailored to their specific needs. However, LLMs often “hallucinate”, generating information that is factually incorrect or incoherent [1]. Examples of hallucinations include stating that the Sun is a planet, or citing nonexistent papers. Teaching students to recognize when AI-generated content warrants independent verification may be the key factor separating AI’s benefits from its potential harms. Needing to evaluate the credibility of AI-generated content may influence how students process and remember information.

Schemas are abstract knowledge structures that organize knowledge and can shape the processing of new information. They typically contain slots that become filled through experience and environmental context [2]. Schemas are crucial to

the learning process, as they shape how new information is encoded and retrieved. In one study, an experimental group that responded to topic-adjacent priming questions before reading a passage significantly outperformed a control group on an immediate memory recall test [3]. In another, Chase and Simon found that chess grandmasters likely have a richer background in topic knowledge than amateur players do, which allows for more robust organization schema, improving encoding and memory recall in chess contexts [4]. Chess grandmasters could almost perfectly reconstruct a chess board after having seen it for only 5 seconds, whereas amateur chess players could only remember the location of several pieces. However, when pieces were placed randomly about the board, amateurs and grand masters performed similarly on memory tests. Chess grand masters have more robust organizational structures that allow them to “chunk” memory into groups instead of encoding them individually. They can recognize certain board states as more meaningful, and encode them as fewer units in memory than amateurs.

Schemas can be activated by a variety of stimuli, not just topic knowledge. For example, source cues can activate schemas and influence memory recall [5]. A preliminary study examined retrieval accuracy between participants exposed to stimuli that contradicted their prior topic knowledge (incongruent schema) and participants exposed to stimuli that aligned with their representations (congruent schema) [6]. The schema incongruent group took longer to respond to questions, but demonstrated better retrieval accuracy. The effect of schema incongruence on memory aligns with other studies demonstrating that skepticism improves recall. For example, Posten and Gino performed a battery of trust-related experiments across 9 studies, and found that trust is associated with both higher acceptance rates of misinformation and reduced memory recall [7]. Both topic knowledge incongruence and source-based skepticism represent cases where unmet expectations may prompt more careful, deliberate processing of information, ultimately improving recall.

Understanding whether labels alone trigger schema activation is important for understanding how students engage with AI-generated educational material. Despite the growing use of

AI-generated content and evidence that source-based schemas influence memory recall, little research has examined whether source attribution to AI-authored text alone is sufficient to activate a source-credibility schema. Therefore, we investigated the effect of an AI-authored label on schema activation, using a memory-based question format to assess recall, and open-ended questions to assess schema activation. Thus, our research question was: "Does labeling an educational passage as AI-generated versus human-authored activate a source-based schema that influences immediate memory recall?"

II. EXPERIMENT DESIGN

We hypothesize that participants in the AI-labeled condition activate a skepticism-associated source credibility schema, resulting in higher immediate memory recall scores compared to participants in the human-authored groups. This hypothesis is based on previously cited data that shows that trust is inversely correlated with recall [7].

We created a survey to assess schema activation and memory recall. The survey had two parts: a short information passage followed by memory recall questions, and several metacognitive questions that asked about the participants' thought process during reading. The metacognitive questions purposefully follow the recall portion to avoid influencing participants' recall scores. We chose a passage topic relatively unknown to most people, but that contained factual information that could easily be tested. Passages were selected from Eric Hobsbawm's *The Age of Revolution* [8] and condensed using a combination of AI-assisted tools and manual editing to improve readability and flow while preserving factual accuracy. Both single-choice and multiple-choice questions were utilized in the survey. The full passage and its memory-related questions can be seen in Appendix A.

Source attribution was altered across three experimental groups. Group 1 was told the passage originated from *The Age of Revolution*, by Eric Hobsbawm [8], Group 2 was told it was summarized from Professor Gregory Clark's *English Industrial Revolution* series on YouTube [9], and Group 3 was told it was generated by ChatGPT when asked to briefly explain several of the key points of the Industrial Revolution in England [10]. Notably, the passage was identical for all three groups. Thus, our independent variable was source attribution, while our dependent variables were participants' scores on the post-passage recall assessment, and schema activation as measured by open-ended metacognitive questions.

The post-recall portion of the survey was designed to assess schema activation and perceived metacognitive outcomes. We asked participants the following questions:

- 1) "What characteristics do you associate with content from this source? List as many descriptors as you would like." (Short answer)
- 2) "Did you have expectations about the passage before reading it? If so, what were they?" (Short answer)
- 3) "The passage was clear and easy to understand." (Likert scale)

- 4) "I believe the information in this passage is accurate." (Likert scale)
- 5) "I feel that I retained the key points from this passage." (Likert scale)
- 6) "While reading this passage, I was attentive to possible inaccuracies." (Likert scale)

Question #1 directly assesses schema activation, while questions #2, #3, and #6 indirectly assess it. Question #4 asks about perceived accuracy of the passage, and question #5 asks about perceived confidence of memory retention.

22 participants were recruited through convenience sampling, including Master's CS students and friends and family of the researcher. Participants were assigned to one of three experimental groups, with 8 in Group 1, 6 in Group 2, and 8 in Group 3.

III. RESULTS

In Table I below, we show group number, number of responses, questions correct, and percentage of questions correct. As a reminder, Group 1 was told the source was *The Age of Revolution* [8], Group 2 a lecture series on YouTube [9], and Group 3 from ChatGPT [10]. Participants in Group 3 scored higher on average than participants in both Groups 1 and 2, indicating potential schema activation.

TABLE I
SURVEY RESPONSES AND SCORES

Group #	# responses	Questions correct	Percentage correct
Group 1	8	5.49/9	61%
Group 2	6	5.37/9	58.7%
Group 3	8	6.99/9	77.7%

To assess schema activation, participants were asked to list characteristics they associated with the source of the passage. Responses were qualitatively grouped into three categories: credibility, academic, and sentiment. We define credibility as how reliable a source is, or how much it can be trusted; academic as how informational or scholarly the source is; and sentiment as how participants felt about the passage. Within each category, we further categorized responses into one of four classifications: positive, neutral, negative, and none. For example, one participant wrote, "Academic, informational, formal language, not too exciting." This would be categorized as positive, positive, negative (for credibility, academic, and sentiment, respectively). Phrasing like "academic" and "informational" implies high levels of credibility, and scholarly writing, while "not too exciting" implies dullness. Responses to question #2 regarding prior expectations were omitted from analysis due to inconsistent results and missing responses.

Table II below shows the results after categorizing responses for the credibility schema. Group 1 perceived their source as credible, as only 1 of 8 participants used phrasing that suggested an unreliable source. Group 2 had no participants say it was unreliable, and 3 of 8 participants in Group 3

thought their source was unreliable. These results suggest that participants generally thought ChatGPT was an unreliable source of information.

TABLE II
CREDIBILITY SCHEMA

Result	Group 1	Group 2	Group 3
positive	4	1	4
neutral	1	3	1
negative	1	0	3
none	2	2	0

Table III below shows the results of the academic schema after categorizing. 5 of 8 participants in Group 1 thought their source was academic, compared with 0 of 6 participants in Group 2, and 3 of 8 in Group 3. Eric Hobsbawm’s book was considered the most academic and credible source, potentially indicating higher levels of trust.

TABLE III
ACADEMIC SCHEMA

Result	Group 1	Group 2	Group 3
positive	5	0	3
neutral	0	1	0
negative	1	1	1
none	2	4	4

Table IV below shows the results of the sentiment schema after categorizing. 2 of 8 participants in Group 1 expressed positive sentiments, and 3 of 8 expressed negative sentiments. In Group 2, 1 of 6 were positive, and 2 of 6 were negative. Lastly, in Group 3, 2 of 8 were positive, and 3 of 8 were negative. Sentiments were roughly even across all three groups, though participants in Group 3 were more likely to have a positive or negative opinion.

TABLE IV
SENTIMENT SCHEMA

Result	Group 1	Group 2	Group 3
positive	2	1	2
neutral	2	2	3
negative	3	2	3
none	1	1	0

Next, we look at metacognitive questions. In Table V, we display perceived clarity of the passage, which was similar across groups. In Group 1, 4 of 8 participants agreed or strongly agreed that the passage was clear and easy to understand, versus 5 of 6 in Group 2, and 4 of 8 in Group 3. No participants in Group 1 and 2 disagreed or strongly disagreed, versus 1 of 8 in Group 3.

TABLE V
THE PASSAGE WAS CLEAR AND EASY TO UNDERSTAND.

Result	Group 1	Group 2	Group 3
Strongly agree	2	0	1
Agree	2	5	3
Neutral	4	1	3
Disagree	0	0	1
Strongly Disagree	0	0	0

In Table VI, we display perceived accuracy of information in the passage. 5 of 8 participants in Group 1 agreed or strongly agreed the information was accurate, compared with 4 in Group 2, and 2 in Group 3. 1 of 8 participants in Group 1 disagreed, versus 0 in Groups 2 and 3. Participants in group 3 were considerably more likely to be skeptical of the accuracy of information.

TABLE VI
I BELIEVE THE INFORMATION IN THIS PASSAGE IS ACCURATE.

Result	Group 1	Group 2	Group 3
Strongly agree	0	1	0
Agree	5	3	2
Neutral	2	2	6
Disagree	1	0	0
Strongly Disagree	0	0	0

In Table VII, we display perceived confidence of memory recall. In Group 1, 5 of 8 participants agreed or strongly agreed that they retained key points from the passage, compared with 3 from Group 2 and 7 from Group 3. Notably, 0 participants from Group 3 responded with neutral, compared to 2 from both Groups 1 and 2. 1 participant from each group disagreed that they retained key points from the passage.

TABLE VII
I FEEL THAT I RETAINED THE KEY POINTS FROM THIS PASSAGE.

Result	Group 1	Group 2	Group 3
Strongly agree	1	0	0
Agree	4	3	7
Neutral	2	2	0
Disagree	1	1	1
Strongly Disagree	0	0	0

Lastly, in Table VIII, we display perceived attention to possible inaccuracies. Only 1 participant in Group 1 agreed or strongly agreed they were aware of possible inaccuracies, compared to 0 in Groups 2, and 2 in Group 3. In addition, Group 3 had fewer disagrees at 3 of 8, compared to both Group 1 and Group 2, which each had 4. Differences across groups were smaller here compared to other measures.

TABLE VIII
WHILE READING THIS PASSAGE, I WAS ATTENTIVE TO POSSIBLE
INACCURACIES.

Result	Group 1	Group 2	Group 3
Strongly agree	1	0	0
Agree	0	0	2
Neutral	3	1	3
Disagree	4	4	3
Strongly Disagree	0	1	0

IV. DISCUSSION

Group 3 scored substantially better than Groups 1 and 2 during recall assessment — nearly 17 percentage points higher. While positive credibility rankings from Groups 1 and 3 were similar, 2 participants from Groups 1 and 2 made no comment about credibility, compared with 0 from Group 3, indicating stronger credibility-related opinions about ChatGPT.

Perceived clarity and accuracy showed similar patterns across groups, but perceived confidence contradicts well-established cognitive science phenomena. Only 2 participants in Group 3 agreed or strongly agreed that the information in the passage was accurate, compared to 5 from Group 1 and 4 from Group 2, with 6 participants in Group 3 neutral, indicating a lack of confidence in accurate information. Participants in Group 3 were more likely to agree that they retained key points from the passage, contradicting cognitive phenomena that indicate confidence is inversely correlated with performance, such as the Dunning-Kruger effect [11].

Taken together, these patterns suggest that exposure to AI-authored sources activates a low-credibility, high-skepticism schema that improves immediate memory recall. This is important for cognitive science, as it further suggests that memory is context dependent and alterable depending on stimuli. Recent cognitive literature has not directly explored the connection between AI-authored and human-authored source biasing. This connection between AI source attribution and schema activation represents an underexplored area that warrants further investigation, particularly as Generative AI becomes more prevalent in educational contexts.

V. CONCLUSION

We found evidence of a potential low-credibility, high-skepticism schema activation in response to AI-authored sources, which may improve immediate recall. Participants in Group 3 scored considerably higher than participants in other groups, and were more likely to be skeptical of the presented information. Future research could improve on this study by increasing the number of participants, increasing diversity of participants, and increasing robustness of both the metacognitive and recall portions of the survey. Our results align with our hypothesis, indicating a potential use of schemas in educational contexts.

Schemas have many applications in cognitive science and in education. In particular, the relationship between source credibility, skepticism, and perceived accuracy has direct implications for how students engage with educational material. Our findings are consistent with previous studies that find an inverse correlation between trust and recall [7]. Additionally, the results of Table VIII further suggest that participants exposed to AI-authored sources were more aware of potential inaccuracies during reading. Therefore, there is a place in curricula for teaching students to be mildly skeptical during reading activities and exercises, as a healthy level of skepticism may deepen understanding of material and improve recall.

VI. LIMITATIONS

While demographic information was not collected, informal observation suggests potential age differences across groups, which may alter scores. The convenience-based nature of sampling additionally contributes to this lack of diversity, as most participants were friends and family of the researcher. The recall assessment could also be extended, but participant recruitment drops as survey length increases. Methodologically, we were unable to measure encoding and depth of processing, instead relying on behavioral inference. Behavior may have been influenced by the Hawthorne Effect, but since all groups would be equally affected, it is unlikely to considerably impact between-group comparisons. Importantly, while patterns emerged from the data, the lack of participants prevents us from conclusively finding evidence to support our hypothesis and answer our research question. Future research should prioritize larger and more diverse samples, and directly measure the effects of encoding and depth of processing to more rigorously test the proposed cognitive mechanism.

VII. RESEARCH PLAN SUMMARY

Total: 115 hours

Week #	Task #	Task description	Estimated hours	Completed (Y/N)
6	1	Review foundational literature	10	Y
7	2	Experiment design	10	Y
7	3	Write passage and quiz questions	10	Y
8	4	Design survey form for section randomization	5	Y
8	5	Pilot testing and design modifications	10	Y
9	6	Recruit participants	30	Y
10	7	Finish reviewing cognitive literature	15	Y
11	8	Data analysis	5	Y
12	9	Writing paper	20	Y
13		TERM PROJECT FINAL REPORT DUE		
13	10	Presentation preparation	5	N
14		TERM PROJECT FINAL PRESENTATION DUE		

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APPENDIX A SURVEY DESIGN

The content in the following passage was taken from The Age of Revolution, by Eric Hobsbawm [8], and summarized using a combination of AI-assisted tools like ChatGPT [10] and manual editing.

A. Passage

Historians generally date the beginning of the Industrial Revolution to the 1780s, rather than earlier decades, because this was when production, trade, and population growth began to increase rapidly and consistently. Economists later described this shift as a "take-off into self-sustained growth," meaning that societies were no longer limited by the production ceilings of pre-industrial economies. Before this change, economic growth was slow and often interrupted by shortages, famine, and technological limitations. By the 1830s and 1840s its social and economic consequences had become widespread. Writers, artists, and social critics increasingly focused on the rise of industrial society, the growth of cities, and the changing nature of work. During this period, new social classes emerged, including the industrial working class, and political movements such as socialism and communism began to gain attention.

Although other European regions experienced industrial development, Britain was uniquely positioned to lead this transformation. Even before industrialization, Britain had higher levels of trade and output per person than most of its competitors. However, Britain's early success was not primarily due to superior science or formal education. In fact, French scientists and engineers were often more advanced, and countries like France and Germany had stronger systems of technical training. Britain lacked a national system of education, and its universities were widely criticized as outdated.

Britain's early industrial growth was fueled by private entrepreneurs and investors responding to market opportunities, especially in industries producing goods for mass consumption, like textiles. The cotton industry was central: it originally grew from overseas trade and imitations of Indian calicoes. Protective policies, such as import prohibitions on Indian cotton, helped British producers establish a domestic market, but the main growth came from exports. Between 1750 and 1769, British cotton exports increased more than tenfold. By 1814, Britain exported about four yards of cotton cloth for every three used domestically, and by 1850, exports reached thirteen yards for every eight used at home.

The Industrial Revolution had significant social consequences, creating discontent among workers, small business owners, and petty-bourgeois groups. Many urban and industrial workers reacted violently, such as smashing machines, an activity associated with the Luddites. These tensions contributed to larger political movements, including the Chartist movement in Britain and the Revolutions of 1848 across Europe. Small business owners and farmers often sympathized with workers, seeing themselves as victims of disruptive industrial changes and profit-driven innovators.

Economically, the Industrial Revolution created inherent risks to profits. Three main challenges were identified: the trade cycle of booms and slumps, a declining rate of profit, and a shortage of profitable investment opportunities. Periodic crises—including unemployment, production drops, and bankruptcies—were common, especially in post-Napoleonic Britain, with notable crises occurring in 1825–26, 1836–37, 1839–42, and 1846–48. However, they were still commonly regarded by businessmen as caused either by particular mistakes—e.g. overspeculation in American stocks—or by outside interference with the smooth operations of the capitalist economy. They were not believed to reflect any fundamental difficulties of the system.

Just as debates over profit, crisis, and capitalism shaped economic life in the early nineteenth century, they also left a deep imprint on cultural production. Between roughly 1789 and 1848, the arts experienced an extraordinary and unusually widespread flourishing across Europe. Artistic achievement was no longer concentrated in a few traditional centers but emerged forcefully in regions such as Russia, Poland, Hungary, and Scandinavia, often alongside growing literacy among the middle classes, which allowed more people to read and engage with literature in their native languages. Literature—especially the novel—and music developed with exceptional intensity, producing figures who still dominate the modern canon, while opera flourished as a popular and politically charged genre across several countries. By contrast, sculpture and architecture were generally less distinguished than in the eighteenth century, with major architectural achievements increasingly produced by engineers rather than artists. Artistic life in this period was profoundly shaped by the dual impact of the French and Industrial Revolutions, which inspired political engagement, social criticism, and new forms of artistic identity. Far from being detached, artists were frequently involved in public affairs, and their work reflected nationalism, social conflict, and the transformation of bourgeois society.

B. Memory Questions

- 1) When do most historians generally date the beginning of the Industrial Revolution?
 - A. Around 1700, when early factories first appeared
 - B. During the 1760s, when trade expanded across Europe
 - C. In the 1780s, when production and growth began rising rapidly
 - D. In the 1830s, when industrial effects spread across Europe
- 2) Which of the following did the passage explicitly reject as the primary reason for Britain's early industrial success?
 - A. Superior scientific knowledge and formal education
 - B. The presence of skilled artisans capable of practical experimentation

- C. Higher levels of trade and output before industrialization
 - D. Market incentives favoring rapid increases in production
 - E. Expanding demand for mass-consumption goods
- 3) According to the passage, which of the following was NOT identified as a threat to profits?
 - A. Alternating phases of growth and recession
 - B. Persistently high costs of raw materials that reduced profit margins
 - C. Limited availability of investments capable of yielding adequate profits
 - D. A long-term downward tendency in returns on capital
 - 4) According to the passage, which of the following interpretations of economic crises was rejected by most businessmen during the early nineteenth century?
 - A. That economic downturns resulted from specific errors such as speculative excess
 - B. That external disruptions could interfere with the normal functioning of markets
 - C. That trade and finance were subject to recurring cycles
 - D. That economic crises reflected inherent structural flaws in the capitalist system
 - E. That peace-time economies could experience severe instability
 - 5) According to the passage, which of the following claims about social conflict during the Industrial Revolution is not supported?
 - A. Worker resistance sometimes took the form of direct attacks on machinery
 - B. Economic change contributed to political mobilization across Europe
 - C. Discontent was shared by industrial workers and some non-industrial groups
 - D. Small business owners and farmers frequently viewed industrial innovators as harmful to their interests
 - E. Sympathy between workers and small proprietors developed into a stable political alliance
 - 6) According to the passage, which of the following statements about the British cotton industry during the Industrial Revolution is correct? (Select all that apply.)
 - A. British cotton exports increased more than tenfold between 1750 and 1769
 - B. Protective measures, such as import prohibitions on Indian cotton goods, helped British producers establish a domestic market
 - C. By 1860, Britain exported about four yards of cotton cloth for every three used domestically
 - D. The cotton industry's early expansion depended primarily on domestic demand rather than exports
 - E. British cotton manufacturing originally developed in connection with overseas trade

- 7) According to the passage, which of the following factors influenced artistic production between 1789 and 1848? (Select all that apply.)
- A. The emergence of a literate middle class as a primary audience for literature and music
 - B. Involvement of artists in public affairs and social criticism
 - C. Direct commissions from a wealthy aristocracy as the primary source of funding
 - D. The impact of the French and Industrial Revolutions
 - E. A widespread focus on sculpture and architecture over literature and music
- 8) According to the passage, which of the following developments are specifically identified as new or especially notable in the arts between 1789 and 1848? (Select all that apply.)
- A. The decline of artistic production in regions such as Russia and Scandinavia
 - B. The continued importance of established Western European artistic centers
 - C. The development of the novel and music, producing figures who remain central to modern literary and musical traditions
 - D. The persistence of Italian painting as a leading artistic tradition
 - E. The unusually wide geographic spread of significant artistic achievement across Europe
- 9) Which of the following sentences appeared verbatim in the passage?
- A. Just as debates over profit, crisis, and capitalism shaped economic life in the early nineteenth century, they also left a deep imprint on cultural production.
 - B. Even prior to industrialization, Britain exhibited higher levels of trade and per capita output than most of its rivals.
 - C. Economically, the Industrial Revolution created inherent risks to profits.
 - D. During this period, new social classes emerged, including the industrial working class, and political movements such as socialism and communism began to gain attention.
 - E. Before this change, economic growth was slow and often interrupted by shortages, famine, and technological limitations.
 - F. As a result, these tensions fed into broader political movements, notably the Chartist movement in Britain and the Revolutions of 1848 throughout Europe.

APPENDIX B

CITI TRAINING CERTIFICATION



Fig. 1. CITI certificate.